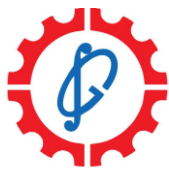


COURSE DESCRIPTION		
Course Code	26GN106 - M	Course Credit: 3
Course Name	Advanced Sensors and Instrumentation	
Course Delivery Mode	TiC	
L-T-P-H	3-0-0-0	
Pre-requisite	Nil	
Year of Introduction and Syllabus Version	2026 V1.0	

COURSE OBJECTIVES		Cognitive Level	Knowledge Category
1	Analyze the structure, elements and characteristics of measurement systems.	L3	K2
2	To evaluate different types of measurement errors using statistical tools and data analysis techniques.	L3	K3
3	To examine the working principles, characteristics and selection criteria of thermal and proximity sensors.	L2	K2
4	To assess the operation and industrial applications of force, pressure, humidity and moisture sensors.	L3	K3

COURSE OUTCOMES (COs)	
After the successful completion of the course students will be able to:	
CO1	Analyze the measurement systems, significance, and their characteristics.
CO2	Evaluate the errors in measurement systems.
CO3	Compare the working principles of different sensors.
CO4	Differentiate the operation of force, pressure, humidity & moisture sensors with their applications.

CO-PO MAPPING						
	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	2	-	2	-	2
CO2	3	3	-	2	-	2
CO3	3	2	2	2	-	2
CO4	3	2	2	2	-	2
Correlation levels: 1- Low; 2-Medium; 3-High; No Correlation- "-"						

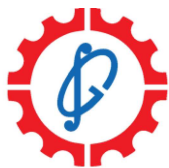


COURSE IMPLEMENTATION CLOCK (IN HOURS)	
Zeroth Module	1 Hr.
Course content delivery hours	36 Hrs.
Internal Test (2 Assessment Tests of one and a half hour duration)	3 Hrs.

COURSE CONTENTS

Zeroth Module: General Introduction to the Course		
- Course introduction with a thorough briefing of Course Objectives and Course Outcomes - An overview of the POs that are mapped to the course	No ESE	1 Hr.

Module-I		COs	Hrs.
Module Title	Measurement Systems		
Introduction - Significance of measurement - Block diagram of measurement system - Methods of measurements - Elements and their functions of measurement systems - Applications - Characteristics of measurement systems - Static and dynamic - Static characteristics - Accuracy - Precision - Sensitivity - Reproducibility - Drift - Static error - Dead zone - Linearity - Resolution - Hysteresis - Loading effects - Dynamic characteristics - Speed of response - Measuring lag - Fidelity - Dynamic error - Calibration.		1	9
Module-II		COs	Hrs.
Module Title	Errors in Measurement and their Statistical Analysis		
Types of error - Gross - Systematic (Instrumental, Environmental, Observational error) - Random error - Statistical treatment of data - Measurement test - Histogram - Arithmetic mean - Dispersion measurement - Range - Deviation - Average deviation - Standard deviation - Variance - Noise - Signal to noise ratio.		2	9
Module-III		COs	Hrs.
Module Title	Thermal & Proximity Sensors		
Introduction - Sensor classifications - Sensor parameters - Selection criterion of sensors - General requirements for interfacing - Temperature sensors - Thermo resistive sensors - Resistance temperature detectors - Thermistor - Thermoelectric sensors - Thermocouple - Electrical sensors - Capacitive position and displacement sensors - LVDT - Proximity sensors - Inductive and capacitive sensors		3	9
Module-IV		COs	Hrs.
Module Title	Force, Pressure, Humidity, and Moisture Sensors		
Force sensor - Strain gauge - Semiconductor strain gauge - Strain gauge accelerometers - Pressure sensors - Mechanical pressure sensors - Piezo resistive pressure sensor - Capacitive pressure sensor - Resistive humidity sensor - Capacitive moisture sensors - Thermal conduction moisture sensors		4	9



TEXT BOOKS

1	Doebelin E. O, and D. N. Manik D. N., <i>Measurement Systems: Application and Design</i> , 6 th Edition, New York, NY, USA: McGraw-Hill Education, 2011, ISBN: 9780070699687.
2	Sawhney A. K., and Sawhney P., <i>A Course in Mechanical Measurements and Instrumentation</i> , 12 th Edition, Dhanpat Rai & Co, 2018, ISBN: 9788177000238.
3	Bentley J. P., <i>Principles of Measurement Systems</i> , 4 th Edition Pearson Education, 2005, ISBN: 9780130430281

REFERENCE BOOKS/ WEB REFERENCES

1	Talbar S. N., Upadhyay A. R., <i>Instrumentation and Measurement</i> , 3 rd Edition, Dhanpat Rai Publishing Company, 2008, ISBN: 9789384378004
2	Johnson C. D., <i>Process Control Instrumentation Technology</i> , 8 th Edition Prentice Hall India, 2013, ISBN: 9788120330290
3	Zeimann V., <i>A Hands-On Course in Sensors Using the Arduino and Raspberry Pi</i> , 2 nd Edition, CRC Press, 2019, ISBN: 9781032376196
4	Fraden J., <i>Handbook of Modern Sensors: Physics, Designs, and Applications</i> , 4 th Edition, Springer, 2014, ISBN: 9781493900404
5	Areny R. P. and Webster J. G., <i>Sensors and Signal Conditioning</i> , 2 nd Edition, Wiley-Interscience, 2000, ISBN: 9780471332329.

MODES OF EVALUATION		SCORE WEIGHTAGE / SPLIT MARKS
Continuous Internal Evaluation (CIE)		50
Internal Test - I		12.5
Internal Test - II		12.5
Taskworks		25
Assignment	50%	
Seminar	50%	
End Semester Examination (ESE)		No ESE
Total		50

APPROVAL SUMMARY

	Number and Date
Recommended by BoS	SCE/BoS/RB/014/25-26, 09.03.2026
Approved by Academic Council	SCE/AC/018(S)/25-26, 24.03.2026